Understanding Miss Rate vs FPPI metric for object detection

Asked 3 years, 4 months ago Modified 3 years, 4 months ago Viewed 5k times

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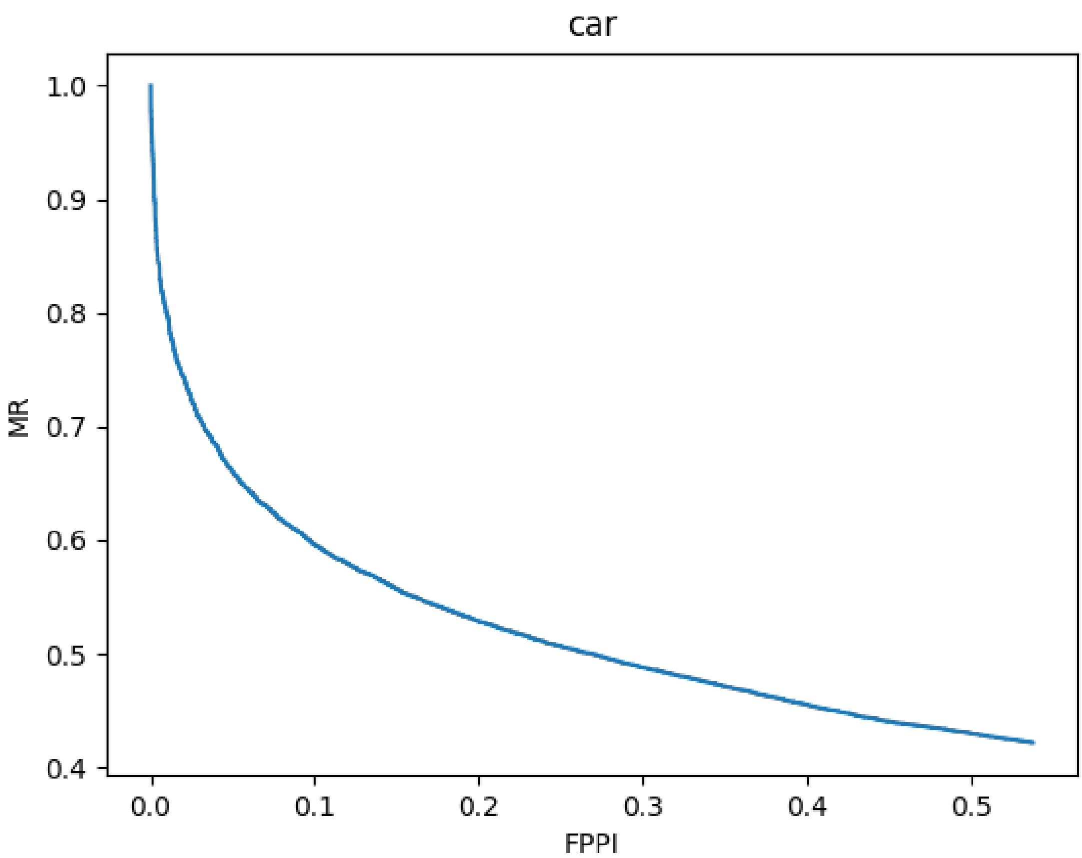
I have a plot between MR vs FPPI for an object detection model which was trained on a dataset for detecting cars. However, I have no clue how to understand this graph and thus am unable to come to a conclusion regarding whether it does a good job or not. I obtained an AP score of 76 which I suppose is pretty good. Can someone please help me understand the graph?



deep-learning computer-vision object-detection false-positive Edit tags

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asked Aug 15, 2019 at 14:33

Jitesh Malipeddi

1,988 2 15 32

1 Answer

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FPPI is short for **False Positive Per Image**. The log-average Miss Rate is a bit similar to Average Precision (the MAP you mentioned) and refers to the objects that are not detected.

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- MR = False Negative / Positive
- FPPI = False Positive / number of tested frames

In this plot, both coordinates are probably log based. This kind of measurement is used when the value of False Positive has an upper limit, no matter how many objects are present in the scene. When you consider a higher threshold for false positive per image, you should expect your miss rate to decrease; since by increasing FPPI, you are accepting more false positives per image and therefore true positives have higher chance of being detected (hence less false negatives). So to sum up, your plot looks fine, as in each image if you want all the objects (cars) to be detected (less Miss Rate), you probably will have some False Positives (objects wrongly detected as car) as a side-effect.

answered Aug 15, 2019 at 16:11
Hadi GhahremanNezhad
2,287 3 23 51